

10 RAG Projects



Real Proof of RAG Expertise

These 10 Projects speak louder than
certificates.

Projects links in caption



Chandra Sekhar
AI Engineer & Educator



PROBLEM

Farmers need quick, reliable information about crop management and pest control without sifting through lengthy documents.



WHAT TO BUILD

A Q&A system where farmers ask natural language questions and receive concise answers from agricultural documents.



HOW IT WORKS

- 1 User asks: "What are common cotton pests and how to manage them?"
- 2 System retrieves relevant snippets from agricultural documents.
- 3 LLM generates a summary with management strategies.



USE CASES

Farmers in the field, agricultural extension workers.



**PROBLEM**

Students struggle to find specific information in textbooks and need a more interactive way to engage with material.

**WHAT TO BUILD**

A system where students ask questions about textbook content and receive answers grounded in the textbook's text.

**HOW IT WORKS**

- 1 User asks: "Explain Newton's first law with an example."
- 2 System retrieves relevant textbook sections.
- 3 LLM provides explanation with examples from the text.

**USE CASES**

Students seeking clarifications, self-study, test prep.



3

Teacher's Lesson Plan Generator



PROBLEM

Teachers need to efficiently create lesson plans aligned with specific textbook topics.



WHAT TO BUILD

A tool where teachers input a textbook and request lesson plans for specific sections or topics.



HOW IT WORKS

- 1 Teacher asks: "Create a lesson plan for Chapter 3 on the Civil War."
- 2 System retrieves relevant sections from the textbook.
- 3 LLM generates lesson plan with objectives, discussion points, and activities.



USE CASES

Teachers preparing daily/weekly lesson plans aligned with textbook content.



Check caption for project guide

4

Enterprise Document Assistant



PROBLEM

Employees struggle to find information in large volumes of internal documents, wasting time.



WHAT TO BUILD

An internal Q&A system where employees ask questions and get answers from company documents.



HOW IT WORKS

- 1 User asks: "What is the paid time off policy?"
- 2 System retrieves relevant sections from HR policy documents.
- 3 LLM provides a summary of the policy.



USE CASES

HR support, IT helpdesk, compliance information, SOPs.



Check caption for project guide



PROBLEM

Researchers need to efficiently extract and synthesize information from numerous academic papers.



WHAT TO BUILD

A system where researchers ask questions about academic papers and receive synthesized answers.



HOW IT WORKS

- 1 User asks: "What are recent trends in deep learning for medical imaging?"
- 2 System retrieves relevant sections from research papers.
- 3 LLM summarizes trends and findings from the papers.



USE CASES

Literature reviews, understanding state-of-the-art, identifying relevant research.





PROBLEM

Villagers lack easy access to information about government schemes and application processes.



WHAT TO BUILD

A system where users ask questions about government schemes and receive step-by-step guidance from official documents.



HOW IT WORKS

- 1 User asks: "How do I apply for PM-KISAN scheme?"
- 2 System retrieves relevant sections from scheme guidelines.
- 3 LLM provides step-by-step application instructions.



USE CASES

Rural populations seeking scheme information and application assistance.





PROBLEM

Small business owners struggle to understand complex legal documents without expensive legal consultation.



WHAT TO BUILD

A system that helps users understand legal documents by answering questions about contracts, terms, and compliance.



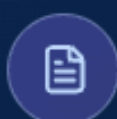
HOW IT WORKS

- 1 User asks: "What are my non-disclosure obligations in this contract?"
- 2 System retrieves relevant clauses from the contract.
- 3 LLM explains obligations and consequences in plain language.



USE CASES

Small business compliance, contract review, tenant rights, employment law.





PROBLEM

Rural populations need immediate health information in their native language during pandemics and disease outbreaks.



WHAT TO BUILD

A multilingual RAG system providing health guidance, symptoms, and prevention measures in local languages from WHO and health ministry documents.



HOW IT WORKS

- 1 User asks: 'What are dengue symptoms?' (in native language)
- 2 System retrieves info from health documents and WHO guidelines.
- 3 LLM provides symptoms, prevention, and when to seek help.



USE CASES

Rural health workers, community programs, pandemic response, disease prevention.





PROBLEM

Entrepreneurs struggle with complex business registration, licensing, and compliance requirements, causing many ideas to never materialize.



WHAT TO BUILD

A comprehensive RAG system guiding users through starting businesses, covering registration, licensing, and compliance for all business types.



HOW IT WORKS

- 1 User asks: "What licenses do I need for a food truck in Hyderabad?"
- 2 System retrieves information from government documents, regulations, and local requirements.
- 3 LLM provides complete checklist with licenses, registrations, and application steps.



USE CASES

Aspiring entrepreneurs, small business owners, startup guidance.



**PROBLEM**

Property owners and contractors face complex building codes and permit requirements that vary by location, risking costly violations.

**WHAT TO BUILD**

A RAG system providing guidance on building codes, permits, construction standards, and approval processes from local and national regulations.

**HOW IT WORKS**

- 1 User asks: "What approvals do I need to add a second floor in Bangalore?"
- 2 System retrieves relevant sections from building bylaws, codes, and approval procedures.
- 3 LLM explains required approvals, setback rules, height restrictions, and documents needed.

**USE CASES**

Homeowners, contractors, architects, developers, building permit offices.



AT A GLANCE

10 Projects. One Skill Tree.

#	PROJECT	DIFFICULTY	DOMAIN
1	Agri Crop Management Q&A	★ Easy	Agriculture
2	Interactive Textbook Q&A	★★ Medium	Education
3	Teacher's Lesson Plan Generator	★★ Medium	Education
4	Enterprise Document Assistant	★★★ Intermediate	Business
5	Academic Research Companion	★★★ Intermediate	Academic
6	Rural Scheme Advisor	★★ Medium	Government
7	Native Language Health Advisory System	★★ Medium	Healthcare
8	Legal Document Navigator	★★★ Intermediate	Legal
9	Business Registration and Startup Guide	★★★ Intermediate	Business
10	Construction and Building Code Advisor	★★★ Intermediate	Construction



Check caption for project guide

Want more content like this?



Tap that follow button and stay in the loop!



Like



Comment



Share



Save



Check caption for project guide